Harrison ("Harry") LaBollita

CONTACT INFORMATION

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+1 (678) 895 - 8180
harrisonlabollita.github.io

RESEARCH

My research interests are primarily centered around strongly correlated electron materials and understanding their macroscopic phenomena (e.g., unconventional superconductivity, strange metal behavior, low-dimensional magnetism, etc.). My work is focused on the application and implementation of state-of-the-art computational tools to study these kinds of materials, from density-functional theory (DFT) to beyond-DFT approaches, like DFT+dynamical mean-field theory (DMFT). These methodologies provide the most comprehensive insights into the electronic structure of strongly correlated materials, and I apply them in a variety of different contexts, including candidate unconventional superconductors, two-dimensional (twisted) magnets, kagomé metals, and other transition-metal compounds.

EMPLOYMENT

Flatiron Institute, Simons Foundation
Flatiron Research Fellow, Center for Computational Quantum Physics — Sept. 2024 –

Arizona State University
Postdoctoral Research Scholar — Dec. 2023 – Sept. 2024

EDUCATION

Arizona State University, Tempe, Arizona — Dec. 2023
Advisor: A. S. Botana; *Doctor of Philosophy (Physics)*

Piedmont University, Demorest, Georgia — May 2019
Bachelor of Science Applied Mathematics & Physics, *Summa Cum Laude*

FELLOWSHIPS & AWARDS

Flatiron Research Fellow — 2024
Postdoctoral fellowship at the Flatiron Institute (a divison of the Simons Foundation) in the Center for Computational Quantum Physics.

Springer Thesis Award — 2024
Awarded by Springer Nature recognizing oustanding Ph.D. theses from around the world in the physical sciences.

Wally Stoelzel Scholarship — 2021 – 2022
Awarded by the Department of Physics at Arizona State to outstanding graduate students.

Teaching Excellence Award — 2020
Awarded by the the Graduate & Professional Student Assocation at Arizona State for graduate students who exemplify excellence in education and classroom instruction.

Arizona State University Summer Graduate Fellowship — 2020
Awarded by the Department of Physics at Arizona State for graduate students to pursue summer research.

NCAA Postgraduate Scholarhsip — 2019
Awarded by the National Collegiate Athletic Assocation (NCAA) to student-athletes who excel academically and athletically.

PUBLICATIONS
⋆ = equal contribution
Citations: 1404
h-index: 17
i10-index: 23

35. "Superconducting phase diagram of multilayer square-planar nickelates",
G. A. Pan, D. F. Segedin, S. F. R. TenHuisen, L. Bhatt, **H. LaBollita**, A. Y. Jiang, Q. Song, A. B. Turkiewicz, D. R. Baykusheva, A. Nag, S. Agrestini, K. Zhou, J. Pelliciari, V. Bisogni, H. Zhou, M. P. M. Dean, H. Paik, D. A. Muller, L. F. Kourkoutis, C. M. Brooks, M. Mitrano, A. S. Botana, B. H. Goodge, J. A. Mundy,
Science 392, 6805 1390 (2026) [arXiv]

34. "Bobkingite, a new coupled sawtooth chain platform",
P. P. Stravropoulos, A. Razpopov, **H. LaBollita**, M. R. Norman, A. S. Botana, R. Valenti

Phys. Rev. Letters (under review) [arXiv]

33. "Squeezing dynamical singlets in bilayer nickelates",
H. LaBollita, A. J. Millis, and O. Gingras,
Phys. Rev. Letters (under review) [arXiv]

32. "Electron vs. hole doping in infinite-layer nickelates: electronic strucutre, magnetism, and correlations",
E. Day-Roberts, F. Bernardini, **H. LaBollita**, Y. Zhao, A. Cano and A. S. Botana,
Phys. Rev. B (under review) [arXiv]

31. "Electronic layer decoupling driven by density-wave order in $La_4Ni_3O_{10}$",
Z. Guan, S. F. R. TenHuisen, M. Tepie, Y. Zhao, E. Day-Roberts, **H. LaBollita**, A. M. Young, X. Cui, X. Chen, F. Glerean, C. A. Guia, M. P. M. Dean, P. Kim, J. F. Mitchell, A. S. Botana, C. C. Homes, and M. Mitrano,
Phys. Rev. Letters 136, 216501 (2026) [arXiv]

30. "Electron-Phonon Origins of Unconventional Resistivity in Moderately Correlated Perovskite Oxides",
J. Coulter, F. B. Kugler, **H. LaBollita**, A. Georges, C. E. Dreyer,
Phys. Rev. Letters **136**, 216301 (2026) [arXiv]

29. "Role of correlations in Ruddlesden-Popper bilayer nickelates under compressive strain",
L. Bleys, N. Corkill, Y. Zhao, G. L. Pascut, **H. LaBollita**, A. S. Botana, and K. F. Quader,
Communication Physics (*under review*) [arXiv]

28. "Lattice-charge coupling in a trilayer nickelate with intertwined density wave order",
X. Jia, Y. Shen, **H. LaBollita**, X. Chen, J. Zhang, Y. Li, H. Zhao, M. G. Kanatzidis, M. Krogstad, H. Zheng, A. Said, A. Alatas, S. Rosenkranz, D. Phelan, M. P. M. Dean, M. R. Norman, J. F. Mitchell, A. S. Botana, and Y. Cao,
Phys. Rev. X **16**, 011013 (2026) [arXiv]

27. "Topotactic oxidation of Ruddlesden-Popper nickelates reveals new structural family: oxygen-intercalated layered perovskites",
D. F. Segedin, J. Kim, **H. LaBollita**, N. K. Taylor, K. Baek, S. H. Sung, A B. Turkiewicz, G. A. Pan, A. Y. Jiang, M. Bambrick-Santoyo, T. Schwaigert, C. K. Kim, A. Tenneti, A. J. Grutter, S. Muramoto, A. T. NDiaye, I. El Baggari, D. A. Walko, H. Zhou, C. M. Brooks, A. S. Botana, D. G. Schlom, J. A. Mundy,
J. Am. Chem. Soc. 148, 5873, 2026 [arXiv]

26. "Unconventional polaronic ground state in superconducting $LiTi_2O_4$",
Z. Hasan, G. A. Pan, **H. LaBollita**, A. Kaczmarek, S. H. Sung, S. Sharma, P. P. Balakrishnan, E. Mercer, V. Bhartiya, Z. Salman, T. Prokscha, A. Suter, A. J. Grutter, M. Garcia-Fernandez, K. Zhou, J. Pelliciari, V. Bisogni, I. El Baggari, D. G. Schlom, M. R. Barone, C. M. Brooks, K. C. Nowack, A. S. Botana, B. D. Faeth, A. de la Torre, and J. A. Mundy,
Nature Communications, **17**, 1303 (2026) [arXiv]

25. "Correlated electronic structure of the alternating single-layer bilayer nickelate $La_5Ni_3O_{11}$",
H. LaBollita and A. S. Botana,
Phys. Rev. B, **113**, 115133 (2026) [arXiv]

24. "Low-temperature transport in high-conductivity correlated metals: A density-functional plus dynamical mean-field study of cubic perovskites,"
H. LaBollita*, J. Lee-Hand*, F. B. Kugler*, L. Van Munoz, J. Kaye, S. Beck, A. Hampel, A. Georges, C. E. Dreyer,
Phys. Rev. B. **113**, 085125 (2026) [arXiv]

23. "Fermi-Liquid T^2 Resistivity: Dynamical Mean-Field Theory Meets Experiment,"
F. B. Kugler*,J. Lee-Hand*, **H. LaBollita***, L. Van Munoz, J. Kaye, S. Beck, A. Hampel, A. Georges, C. E. Dreyer,
Phys. Rev. B **113**, L081105 (2026) [arXiv]

22. "Stabilizing the calculation of the self-energy in dynamical mean-field theory using constrained residual minimization",
H. LaBollita, J. Kaye, and A. Hampel,

Phys. Rev. B **111**, 115155 (2025) [arXiv]

Before CCQ

21. "Assessing spin-density wave formation in $La_3Ni_2O_7$ from electronic structure calculations,"
H. LaBollita, V. Pardo, M. R. Norman, and A. S. Botana,
Phys. Rev Materials **8**, L111801 (2024) [arXiv]

20. "Electronic correlations, layer distinction, and electron doping in the alternating single-layer trilayer $La_3Ni_2O_7$ polymorph,"
H. LaBollita, S. Bag, J. Kapeghian, and A. S. Botana,
Phys. Rev B **110**, 155145 (2024) [arXiv]

19. "Pressure effects on the electronic structure and magnetic properties of infinite-layer nickelates,"
S. Sharma, M. C. Jung, **H. LaBollita**, and A. S. Botana,
Phys. Rev. B **110**, 155156 (2024) [arXiv]

18. "Electronic structure and magnetic tendences of trilayer $La_4Ni_3O_{10}$ under pressure: structural transition, molecular orbitals, and layer differentiation,"
H. LaBollita, J. Kapeghian, M. R. Norman, and A. S. Botana,
Phys. Rev. B **109**, 195151 (2024) [arXiv]

17. "Electronic band structure of a superconducting nickelate probed by the seebeck coefficient in the disordered limit,"
G. Grissonnanche, G. A. Pan, **H. LaBollita**, D. Ferenc Segedin, Q. Song, H. Paik, C. M. Brooks, A. S. Botana, J. A. Mundy, and B. J. Ramshaw
Phys. Rev. X **14**, 041021 (2024) [arXiv]

16. "Capping effects on spin and charge excitations in parent and superconducting $Nd_{1-x}Sr_xNiO_2$,"
S. Fan, **H. LaBollita**, Q. Gao, N. Khan, Y. Gu, T. Kim, J. Li, V. Bhartiya, Y. Li, W. Sun, J. Yang, S. Yan, A. Barbour, X. Zhou, A. Cano, F. Bernardini, Y. Nie, Z. Zhub, V. Bisogni, C. Mazzoli, A. S. Botana, and J. Pelliciari,
Phys. Rev. Lett **133**, 206501 (2024) [arXiv]

15. "Hydrogen is not necessary for superconductivity in topotactically reduced nickelates,"
P. P. Balakrishnan, D. Ferenc Segedin, L. E. Chow, P. Quarterman, S. Muramoto, M. Surendran, R. K. Patel, **H. LaBollita**, G. A. Pan, Q. Song, Y. Zhang, I. E. Baggari, K. Jagadish, Y. Shao, B. H. Goodge, L. F. Kourkoutis, S. Middey, A. S. Botana, J. Ravichandran, A. Ariando, J. A. Mundy, and A. J. Grutter,
Nature Communications **15**, 7387 (2024) [arXiv]

14. "Polymorphism in Ruddlesden-Popper $La_3Ni_2O_7$: Discovery of Hidden Phase with Distinctive Layer Stacking,"
X. Chen, J. Zhang, A. S. Thind, S. Sharma, **H. LaBollita**, G. Peterson, H. Zheng, D. P. Phelan, A. S. Botana, R. F. Klie, and J. F. Mitchell,
J. Am. Chem. Soc. 146, 3640 [arXiv]

13. "Conductivity of infinite-layer $NdNiO_2$ as a probe of spectator bands,"
H. LaBollita, A. Hampel, J. Karp, A. S. Botana, and A. J. Millis,
Phys. Rev. B **107**, 205155 (2023) [arXiv]

12. "Limits to the strain engineering of layered-square planar nickelate thin films,"
D. Ferenc Segedin, B. H. Goodge, G. A. Pan, Q. Song, **H. LaBollita**, M. Jung, H. El-Sherif, S. Doyle, A. Turkiewicz, N. K. Taylor, J. A. Mason, A. T. N'Diaye, H. Paik, I. E. Baggari, A. S. Botana, L. F. Kourkoutis, C. M. Brooks, and J. A. Mundy,
Nature Communications **14**, 1468 (2023) [arXiv]

11. "Orbital polarization, charge transfer, and fluorescence in reduced valence nickelates,"
M. R. Norman, A. S. Botana, J. Karp, A. Hampel, **H. LaBollita**, A. J. Millis, G. Fabbris, Y. Shen, and M. P. M. Dean,
Phys. Rev. B **107**, 165124 (2023) [arXiv]

10. "Antiferromagnetic metal phase in an electron doped rare earth nickelate,"
Q. Song, S. Doyle, G. A. Pan, I. E. Baggari, D. Ferenc Segedin, D. C. Carrizales, J. Nordlander, C. Tzscaschel, J. R. Ehrets, Z. Hasan, H. El-Sherif, J. Krishna, C. Hanson, **H. LaBollita**, A. Bostwick, C. Jozwiak, E.

Rotenberg, S. Xu, A. Lanzara, A. N'Diaye, C. A. Heikes, Y. Liu, H. Paik, C. M. Brooks, B. Pamuk, J. T. Heron, P. Shafer, W. D. Ratcliff, A. S. Botana, L. Moreschini, and J. A. Mundy,
Nature Physics **19** 522-528 (2023) [arXiv]

9. "Many-body electronic structure of $d^{9-\delta}$ nickelates,"
H. LaBollita, M. Jung, and A. S. Botana,
Phys. Rev. B **106**, 115132 (2022) [arXiv]

8. "Antiferromagnetic insulating state in layered nickelates at half filling,"
M. Jung, **H. LaBollita**, V. Pardo, and A. S. Botana,
Scientifc Reports **12**, 17864 (2022) [arXiv]

7. "Visualizing the out-of-plane electronic correlations in an intercalated transition metal dichalcogenide,"
X. Xiang*, **H. LaBollita***, Z. Cheng, H. Bhandari, T. A. Cochran, J. Yin, M. S. Hossain, I. Belopolski, Q. Zhang, Y. Jiang, N. Shumiya, D. Multer, M. Liskevich, D. A. Usanov, Y. Dang, V. N. Strocov, A. V. Davydov, N. J. Ghimire, A. S. Botana, and M. Z. Hasan,
Phys. Rev. B **105**, L121107 (2022) [arXiv]

6. "Correlated electronic structure of a quintuple-layer nickelate,"
H. LaBollita and A. S. Botana,
Phys. Rev. B **105** 085118 (2022) [arXiv]

5. "Superconductivity in a quintuple-layer square-planar nickelate,"
G. A. Pan, D. Ferenc Segedin, **H. LaBollita**, Q. Song, E. M. Nica, B. H. Goodge, A. T. Pierce, S. Doyle, S. Novakov, D. C. Carrizales, A. T. N'Diaye, P. Shafter, H. Paik, J. T. Heron, J. A. Mason, A. Yacoby, L. F. Kourkoutis, O. Erten, C. M. Brooks, A. S. Botana, and J. A. Mundy,
Nature Materials 21, 160-164 (2022) [arXiv]

4. "Tuning the Van Hove singularities in AV_3Sb_5 (A = K, Rb, Cs) via pressure and doping,"
H. LaBollita and A. S. Botana,
Phys. Rev. B **104**, 205129 (2021) [arXiv]

3. "Moiré skyrmions and chiral magnetic phases in twisted CrX_3 (X = I, Br, Cl) bilayers,"
M. Akram*, **H. LaBollita***, D. Dey, J. Kapeghian, A. S. Botana, and O. Erten,
Nano Letters 21, 15, 6633-6639 (2021) [arXiv]

2. "Electronic structure and magnetic properties of higher-order nickelates: $La_{n+1}Ni_nO_{2n+2}$ ($n = 4 - 6$),"
H. LaBollita and A. S. Botana,
Phys. Rev. B **104** 035148 (2021) [arXiv]

1. "Effects of Sr-doping on the electronic and spin-state properties of infinite-layer nickelates: Nature of holes,"
J. Krishna, **H. LaBollita**, A. O. Fumega, V. Pardo, and A. S. Botana,
Phys. Rev. B **102**, 224506 (2020) [arXiv]

INVITED TALKS

"The nickel age of high-temperature superconductivity"
Institute of Theoretical Physics, University of Cologne, Cologne, Germany — Jul. 2026

"The nickel age of high-temperature superconductivity"
Hunter Physics Colloquium, Hunter Physics, New York, NY, USA — Apr. 2026

"Electronic structure of Ruddlesden-Popper nickelates"
APS Global Summit, Denver, CO, USA — Mar. 2026

"Square-planar nickelates: where nickelate superconductivty began"
University of Oklahoma Physics and Astronomy Colloquium, Norman, Oklahoma, USA — Feb. 2026

"From square-planar to octahedral nickelates: an electronic structure perspective"
Frontiers of Superconductivity Workshop by eQMA, Online — Feb. 2026

"ModEST: A generic framework for electronic structure developments in TRIQS"

TRIQS Developer Meeting, Collège de France, Paris, France — Jun. 2025

"Electronic correlations in Ruddlesden-Popper nickelates"
Brookhaven National Laboratory, Upton, NY, USA — May 2025

"Nickelate superconductivity to infinity and beyond"
University of Geneva, Geneva, Switzerland — May 2025

"Electronic structure of superconducting nickelates: from infinity and beyond"
Gordon Research Seminar, Vaud, Switzerland — May 2025

"Electronic structure of Ruddlesden-Popper nickelates"
APS Global Summit, Anaheim, CA, USA — Mar. 2025

"Superconducting nickelates: reduced, parents, and polymorphs"
Gli Amici del QMAC, Paris, France — Jan. 2025

"Can we do away with tail-fitting in TRIQS"
TRIQS Developer Meeting, Collège de France, Paris, France — Sept. 2023

WORKSHOPS & CONFERENCES

M^2S Materials and Mechanisms of Superconductivity, Stuggart, Germany	Jul. 2026
American Physical Society, Global Summit, Denver, CO, USA	Mar. 2026
American Physical Society, Global Summit, Anaheim, CA, USA	Mar. 2025
American Physical Society, March Meeting, Minneapolis, MN, USA	Mar. 2024
American Physical Society, March Meeting, Las Vegas, NV, USA	Mar. 2023
American Physical Society, March Meeting, Chicago, IL, USA	Mar. 2022
American Physical Society, March Meeting, Online due to COVID-19	Mar. 2021
American Physical Society, March Meeting, Boston, MA, USA	Mar. 2019

MENTORING

Interns

Christine Zhang (Flatiron CCQ)
Undergraduate, Mathematics & Physics, Columbia University

Ethan Dehal (SCENE, Arizona State University)
High-school student

Adriana Baniecki (SCENE, Arizona State University) — Spring 2022
now Undergraduate, Physics & Math, Notre-Dame University

Siva Bubby (SCENE, Arizona State University) — Spring 2021
now Undergraduate, Case & Western University

TEACHING

Arizona State University, Tempe, AZ

Teaching Assistant, PHY 121: Mechanics for Engineers	Spring 2020
Teaching Assistant, PHY 131: Electricity & Magnetism for Engineers	Fall 2019, Fall 2020

PROFESSIONAL ACTIVITIES

Department service

Co-organizer, DMFT-Internal Meeting, *CCQ, Flatiron Institute*	2024 –
Co-organizer, Grad2Grad Talks, ASU Department of Physics	2021 – 2023
Graduate Student Representative, ASU Department of Physics Bylaws Committee	2021 – 2022

Mentor, ASU Sundial Project 2020 – 2021

Outreach

Instructor, Clubes de Ciencia 2021
Organizer, Maker Faire, *Henry Ford Museum* 2018
Organizer, UF Center for Pre-Collegiate Education and Training, *University of Florida* 2017

Journal refereeing

Physical Review Letters (APS) 2023 –
Physical Review B (APS) 2023 –
Nature Communications 2023 –
Nature 2025 –